Towards Robust 3D Plant Phenotyping: Robotic RGB Reconstruction, Point Cloud Segmentation, and Self-Supervised Adaptation

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Abstract

Robust 3D plant phenotyping requires accurate plant reconstruction and reliable analysis of reconstructed point clouds. This paper presents a robotic pipeline using a monocular RGB camera mounted on a 7-axis JetCobot arm for automated plant acquisition. We evaluate state-of-the-art 3D reconstruction methods, including COLMAP and MASt3R, combined with NeRF and 3D Gaussian Splatting, and quantify reconstruction quality against ground-truth data. The reconstructed point clouds are then evaluated with multiple 3D segmentation methods to measure the impact of reconstruction artifacts and point dropout on plant structure understanding. To improve robustness across clean and reconstructed point clouds, we further investigate self-supervised representation learning using Barlow Twins and compare it with Utonia. The results provide an end-to-end evaluation of robotic 3D plant acquisition, reconstruction, segmentation, and domain adaptation, highlighting the challenges of reconstruction-induced degradation and the potential of self-supervised learning for robust automated plant phenotyping.

Index Terms— *3D plant phenotyping, 3D reconstruction, Barlow Twins, point cloud segmentation, robotic vision, self-supervised learning.*

1. Introduction

Plant phenotyping requires accurate, non-invasive, and repeatable measurements of plant structure [1]. Conventional measurements are time-consuming, while 2D imaging is limited by leaf overlap, self-occlusion, and the lack of depth information [2]. Monocular RGB cameras provide a low-cost alternative for 3D plant reconstruction.

This work presents a robotic pipeline for automated multi-view plant acquisition using a monocular RGB camera mounted on a 7-axis collaborative robotic arm. ROS 2 [3] and MoveIt2 [4] enable robotic scanning, while COLMAP [5] and MASt3R [6] are evaluated for 3D reconstruction with NeRF [7] and 3D Gaussian Splatting [8]. The reconstructed point clouds are further evaluated for semantic segmentation and plant structure analysis, including a self-supervised representation learning stage based on Utonia [9] and Barlow Twins [10] to investigate robustness to reconstruction noise.

The main contributions are:

- A ROS 2-based pipeline for automated multi-view plant acquisition [3].
- An evaluation of COLMAP and MASt3R for monocular 3D reconstruction [5], [6].
- Evaluation of 3D segmentation under clean and reconstructed point-cloud conditions [11]–[13].
- Self-supervised representation learning with Utonia and Barlow Twins for robust 3D plant analysis [9], [10].
- Integration of the reconstructed plant data into the **PheNo** phenotyping infrastructure.

The remainder of the paper presents the system, reconstruction methods, segmentation experiments, and self-supervised learning results.

2. Related Work

Robotic platforms enable automated multi-view plant acquisition [14]. For monocular 3D reconstruction, COLMAP [5] provides Structure-from-Motion pose estimation, while MASt3R [6] enables learning-based image matching and 3D reconstruction. Their outputs can be used with NeRF [7] and 3D Gaussian Splatting [8] for 3D scene representation.

3D point-cloud segmentation enables plant organ analysis and phenotyping [15], but reconstruction noise can degrade segmentation performance. Self-supervised learning offers a way to learn more robust 3D representations. This work evaluates Utonia [9] as a baseline and investigates Barlow Twins [10] for aligning representations between clean and reconstructed plant point clouds.

The proposed pipeline combines robotic acquisition, 3D reconstruction, segmentation, and self-supervised representation learning for robust plant phenotyping.

3. System Overview

3.1 Robot Platform

The acquisition system is built around the **JetCobot**, a 7-axis collaborative robotic arm manufactured by Elephant Robotics, chosen for its flexible positioning within a compact workspace suitable for benchtop plant scanning. The arm is powered by an NVIDIA Jetson Orin Nano, enabling on-board processing, while a monocular RGB camera is mounted at the wrist for close-range imaging. Figure 1 shows the physical JetCobot platform used in this work.

Fig. 1: The JetCobot 7-axis collaborative robotic arm.

3.2 ROS 2 Software Architecture

The software stack is built on **ROS 2 Jazzy Jalisco** [3], which provides the middleware for communication and control between the perception, planning, and actuation modules. **MoveIt2** [4] is configured with the JetCobot's kinematic chain to plan collision-free trajectories for the hemispherical scanning motion described in Section 4. Prior to hardware deployment, the trajectories are validated in a Gazebo simulation environment using photorealistic plant assets created in Blender. This sim-to-real workflow allows scan trajectories to be verified for reachability and collision safety before execution on the physical robot, while the simulation also provides ground-truth camera poses for evaluation. Figure 2 illustrates the simulated and physical scanning setups.

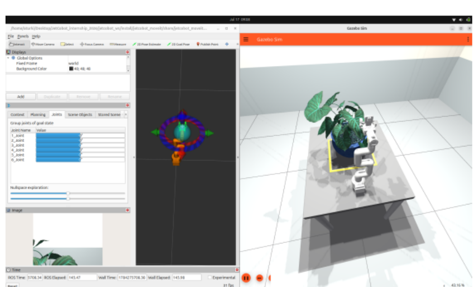


Fig. 2: The RViz/MoveIt2 motion-planning view, the Gazebo simulation environment, and the physical scanning setup.

4. Methodology

4.1 Data Capture Pipeline

During scanning, RGB images are published through ROS 2 while the wrist-mounted camera follows a hemispherical trajectory around the plant. The end-effector pose is obtained from forward kinematics and synchronized with each image using a common timestamp. The resulting image–pose pairs are used for reconstruction.

Figure 3 illustrates the data-capture pipeline.

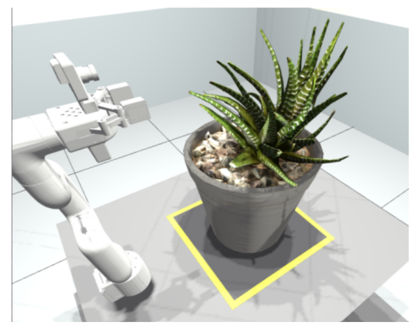


Fig. 3: RGB image and forward-kinematic pose acquisition through ROS 2.

4.2 3D Reconstruction Back Ends

Two reconstruction back ends are evaluated:

- **COLMAP** [5]: classical Structure-from-Motion for camera pose estimation and sparse reconstruction.
- **MASt3R** [6]: learning-based dense matching and 3D reconstruction from RGB image pairs.

The recovered poses initialize two neural scene representations: **NeRF** [7] and **3D Gaussian Splatting (3DGS)** [8]. NeRF is evaluated with COLMAP and MASt3R poses, denoted NeRF (COLMAP) and NeRF (MASt3R), while 3DGS uses the COLMAP reconstruction, denoted GS (COLMAP).

Figure 4 compares the resulting point clouds.

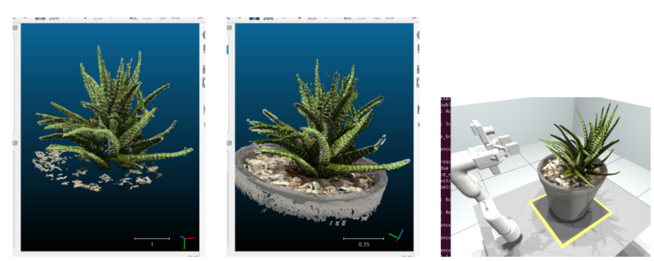


Fig. 4: Reconstructed point clouds from COLMAP and MASt3R compared with the ground-truth plant.

4.3 Evaluation Protocol

Reconstruction quality is evaluated using PSNR, SSIM, LPIPS, and rendering speed (fps), while camera-pose accuracy is evaluated against the Blender ground-truth trajectory using translation ATE-RMSE and rotation error.

For the pose evaluation, let N denote the total number of corresponding camera poses shared by the ground-truth and estimated trajectories, with $i = 1, \ldots, N$ indexing the matched poses. Let $\mathbf{t}_i \in \mathbb{R}^3$ and $\mathbf{R}_i \in SO(3)$ denote the ground-truth translation and rotation at the i-th camera pose, respectively. The corresponding estimated translation and rotation obtained from the reconstruction method are denoted by $\hat{\mathbf{t}}_i \in \mathbb{R}^3$ and $\hat{\mathbf{R}}_i \in SO(3)$, respectively. The symbol $\hat{(\cdot)}$ indicates an estimated quantity, while the absence of a hat denotes the ground-truth quantity.

For the N matched camera poses, the Absolute Trajectory Error (ATE) in translation is quantified using the root mean square error (RMSE):

$$\mathrm{ATE}_{\mathrm{RMSE}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{t}_i - \hat{\mathbf{t}}_i\right\|_2^2}. \tag{1}$$

The rotation error for the i-th pose is computed as the angular difference between the ground-truth and estimated rotation matrices:

$$e_R^{(i)} = \cos^{-1}\left(\frac{\mathrm{tr}\left(\mathbf{R}_i^T\hat{\mathbf{R}}_i\right) - 1}{2}\right)\frac{180}{\pi}, \tag{2}$$

where $\mathrm{tr}(\cdot)$ denotes the matrix trace and $e_R^{(i)}$ is expressed in degrees. The mean rotation error over all matched poses is then reported as

$$\bar{e}_R = \frac{1}{N}\sum_{i=1}^{N} e_R^{(i)}. \tag{3}$$

5. Reconstruction Evaluation Results

5.1 Rendering Quality

Table I summarizes the rendering-quality metrics obtained for the three evaluated pipelines. NeRF (COLMAP) achieves the highest PSNR (19.93 dB), indicating superior overall reconstruction fidelity, while GS (COLMAP) achieves the best SSIM (0.85) and LPIPS (0.11), reflecting the sharper, more perceptually faithful renders typical of Gaussian Splatting. NeRF (MASt3R) trails both COLMAP-based pipelines across all quality metrics, which we attribute to the pose-estimation errors discussed in Section 5.2.

TABLE I: Reconstruction Quality Metrics Across Pipelines.

Metric	NeRF (COLMAP)	NeRF (MASt3R)	GS (COLMAP)
PSNR ↑	**19.93**	14.17	19.00
SSIM ↑	0.79	0.55	**0.85**
LPIPS ↓	0.16	0.42	**0.11**

5.2 Camera Pose Accuracy

Figure 5 compares the absolute trajectory error (ATE) and rotation error of COLMAP and MASt3R against the Blender ground truth. COLMAP maintains low, consistent errors throughout the scan. Conversely, MASt3R exhibits severe initial error spikes caused by pairwise drift under repetitive leaf textures.

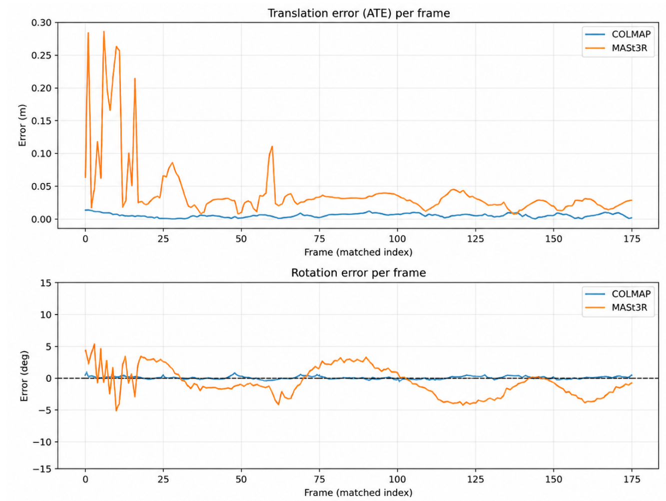
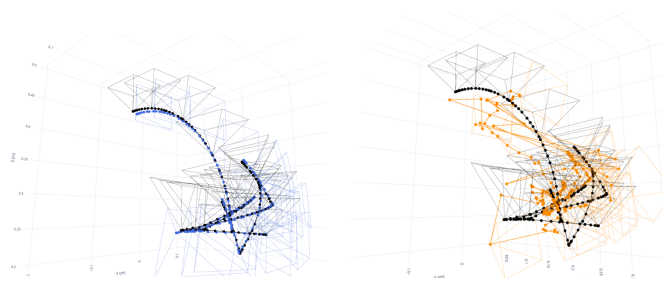


Fig. 5: Per-frame translation (ATE) and rotation errors relative to Blender ground truth.



(a) COLMAP(blue)vs. GT (black)

(b) MASt3R(orange) vs. GT(black)

Fig. 6: 3D camera trajectories overlaid on the ground-truth path.

6. 3D Plant Segmentation

The reconstructed plant point clouds are evaluated for semantic and organ-level segmentation. To assess the effect of reconstruction noise, the segmentation models are applied to both clean ground-truth point clouds and reconstructed point clouds.

6.1 Segmentation Models

Three 3D point-cloud segmentation models are evaluated: PointNet++ [11], DGCNN [12], and Point Transformer V3 (PTv3) [13]. These models provide different approaches to learning local and global geometric representations from 3D point clouds.

6.2 Experimental Setup

The plant used for the segmentation experiments is obtained from the **PLANesT-3D** dataset [16], which provides annotated 3D plant point clouds with manually labeled leaf and stem classes and organ instances. These annotations are used as ground truth for training and evaluation.

Three models are evaluated: PointNet++, DGCNN, and Point Transformer V3 (PTv3). The models are trained using the clean ground-truth point cloud and evaluated under two conditions: the original ground-truth point cloud and the corresponding reconstructed point clouds. The latter contain reconstruction artifacts such as missing points, uneven density, and geometric and appearance variations, providing a realistic measure of segmentation robustness to the reconstruction process.

For a given class, the Intersection over Union (IoU) is defined as

$$\mathrm{IoU} = \frac{TP}{TP + FP + FN}, \tag{4}$$

where TP, FP, and FN denote the numbers of true-positive, false-positive, and false-negative points, respectively.

The mean Intersection over Union (mIoU) over C classes is defined as

$$\mathrm{mIoU} = \frac{1}{C}\sum_{c=1}^{C} \mathrm{IoU}_c, \tag{5}$$

where C denotes the total number of evaluated classes, c is the class index, and IoU_c denotes the IoU for class c.

6.3 Segmentation Results

Table II reports the baseline performance obtained on the ground-truth point cloud. PTv3 achieves the highest mIoU (0.9354), followed by PointNet++ (0.9064) and DGCNN (0.6205). These results provide the reference for evaluating segmentation performance on the reconstructed point clouds.

TABLE II: Baseline 3D plant segmentation performance on the ground-truth point cloud.

Model	mIoU	Accuracy
PointNet++	0.9064	**0.9886**
PTv3	**0.9354**	0.9829
DGCNN	0.6205	0.7652

PTv3 is selected as the best-performing model according to mIoU. Figure 7 shows its semantic and instance segmentation results on the ground-truth plant.

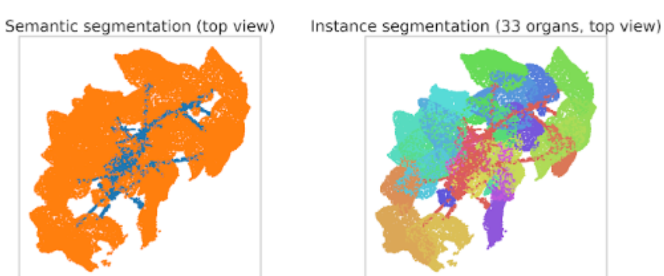


Fig. 7: PTv3 segmentation results on the ground-truth plant: semantic segmentation (left) and instance segmentation (right).

The performance on reconstructed point clouds is subsequently compared with the ground-truth baseline using the mIoU degradation defined in Section 6.2.

6.4 Degradation on Reconstructed Point Clouds

The robustness of each segmentation model is evaluated by comparing its performance on the ground-truth and reconstructed point clouds. The mIoU degradation is defined as

$$\Delta_{\mathrm{mIoU}} = \mathrm{mIoU}_{\mathrm{GT}} - \mathrm{mIoU}_{\mathrm{reconstructed}}, \tag{6}$$

with the relative degradation given by

$$D_{\mathrm{mIoU}} = \frac{\Delta_{\mathrm{mIoU}}}{\mathrm{mIoU}_{\mathrm{GT}}} \times 100. \tag{7}$$

A larger positive value indicates a greater loss of segmentation performance after reconstruction.

Table III summarizes the degradation observed for the evaluated models. The results are used to identify the model most affected by reconstruction artifacts and to motivate the self-supervised representation learning experiments presented in Section 7.

TABLE III: segmentation performance across reconstructed point-cloud.

Model	GT mIoU	Reconstructed mIoU	Degradation (%)
PointNet++	0.9064	0.8776	3.18
DGCNN	0.6205	0.5916	4.66
PTv3	0.9354	0.8731	6.65

7. Self-Supervised Learning

The results in Section 6 show that reconstruction artifacts degrade segmentation performance. Self-supervised learning is therefore used to align representations of clean and reconstructed point clouds of the same plant without additional annotations.

7.1 SSL Setup

Point Transformer v3 (PTv3) serves as the feature backbone $f_\theta : \mathbb{R}^{N\times(3+D)} \to \mathbb{R}^{N\times D_h}$, where N denotes the number of points, D denotes the number of point attributes in addition to the three spatial coordinates, and D_h denotes the dimensionality of the point-level feature representation.

Before feature extraction, both the ground-truth and reconstructed point clouds are sampled to the same number of points N to ensure compatible input dimensions for the shared backbone.

Given a batch of B paired point clouds, each pair consists of a clean ground-truth point cloud and its corresponding reconstructed (distorted) version. The two views are denoted by $\mathbf{X}_{\mathrm{GT}}^{(b)} \in \mathbb{R}^{N\times(3+D)}$ and $\mathbf{X}_{\mathrm{REC}}^{(b)} \in \mathbb{R}^{N\times(3+D)}$, where $b = 1, \ldots, B$ indexes the paired samples in the batch. The shared backbone f_θ extracts point-level feature representations for both views:

$$\mathbf{H}_{\mathrm{GT}}^{(b)} = f_\theta\left(\mathbf{X}_{\mathrm{GT}}^{(b)}\right), \tag{8}$$

$$\mathbf{H}_{\mathrm{REC}}^{(b)} = f_\theta\left(\mathbf{X}_{\mathrm{REC}}^{(b)}\right), \tag{9}$$

where $\mathbf{H}_{\mathrm{GT}}^{(b)}$ and $\mathbf{H}_{\mathrm{REC}}^{(b)} \in \mathbb{R}^{N\times D_h}$ denote the corresponding point-level feature matrices.

For each point cloud, global average pooling GAP$(\cdot)$ aggregates the point features along the point dimension N, producing a cloud-level representation in $\mathbb{R}^{D_h}$. A shared multi-layer perceptron projection head $g_\phi : \mathbb{R}^{D_h} \to \mathbb{R}^{D_z}$ then maps each representation into a D_z-dimensional embedding space:

$$\mathbf{z}_{\mathrm{GT}}^{(b)} = g_\phi\left(\mathrm{GAP}\left(\mathbf{H}_{\mathrm{GT}}^{(b)}\right)\right), \tag{10}$$

$$\mathbf{z}_{\mathrm{REC}}^{(b)} = g_\phi\left(\mathrm{GAP}\left(\mathbf{H}_{\mathrm{REC}}^{(b)}\right)\right), \tag{11}$$

where D_z denotes the dimensionality of the embedding space and $\mathbf{z}_{\mathrm{GT}}^{(b)}, \mathbf{z}_{\mathrm{REC}}^{(b)} \in \mathbb{R}^{D_z}$ denote the embeddings of the b-th ground-truth and reconstructed point clouds, respectively. The embeddings are subsequently stacked across the B paired samples to form the batch embedding matrices $\mathbf{Z}_{\mathrm{GT}}, \mathbf{Z}_{\mathrm{REC}} \in \mathbb{R}^{B\times D_z}$.

7.2 Barlow Twins

Barlow Twins [10] aligns representations by decorrelating feature dimensions without relying on negative samples. Let $\widetilde{\mathbf{Z}}_{\mathrm{GT}}, \widetilde{\mathbf{Z}}_{\mathrm{REC}} \in \mathbb{R}^{B\times D_z}$ denote the embeddings normalized along the batch dimension (zero mean, unit variance). The cross-correlation matrix $\mathbf{C} \in \mathbb{R}^{D_z\times D_z}$ between representations is defined as:

$$\mathbf{C} = \frac{1}{B}\widetilde{\mathbf{Z}}_{\mathrm{GT}}^{T}\widetilde{\mathbf{Z}}_{\mathrm{REC}}, \tag{12}$$

where $C_{ij} \in [-1, 1]$ measures the correlation between feature dimension i of the ground-truth cloud and feature dimension j of the reconstructed cloud. The Barlow Twins loss function $\mathcal{L}_{\mathrm{BT}}$ drives $\mathbf{C}$ toward the identity matrix $\mathbf{I}$:

$$\mathcal{L}_{\mathrm{BT}} = \sum_{i=1}^{D_z}(1 - C_{ii})^2 + \lambda \sum_{i=1}^{D_z}\sum_{j\neq i}^{D_z} C_{ij}^2, \tag{13}$$

where the invariance term $\sum_i (1 - C_{ii})^2$ enforces distortion robustness by setting diagonal elements to 1, while the redundancy reduction term $\sum_i \sum_{j\neq i} C_{ij}^2$ decorrelates off-diagonal entries. The hyperparameter $\lambda > 0$ balances redundancy reduction against invariant feature learning.

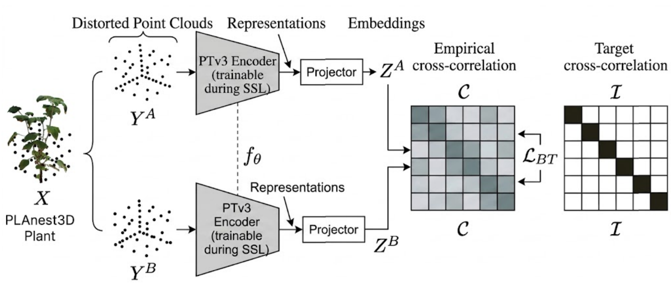


Fig. 8: Barlow Twins self-supervised pipeline for aligning cloud-level representations $\mathbf{Z}_{\mathrm{GT}}$ and $\mathbf{Z}_{\mathrm{REC}}$ toward target cross-correlation identity matrix $\mathbf{I}$.

7.3 Comparison With Utonia

Barlow Twins is compared with Utonia [9] using the same PTv3 backbone, input data, and downstream segmentation protocol. The evaluated configurations are:

1) supervised PTv3;
2) PTv3 + Utonia;
3) PTv3 + Barlow Twins.

7.4 Training Procedure

The PTv3 backbone is initialized from supervised training on the clean ground-truth point cloud. Paired clean and reconstructed samples are then processed by the shared encoder and optimized using the Barlow Twins loss. The projection head is removed before downstream segmentation evaluation. The main parameters are reported in Table IV.

TABLE IV: Self-supervised training configuration.

Parameter	Barlow Twins	Utonia
SSL epochs	300	40
Batch size	32	8
Learning rate	1×10^{-4}	1×10^{-3}
Weight decay	1×10^{-6}	1×10^{-4}
Optimizer	AdamW	AdamW
Warm-up epochs	10	–
λ	5×10^{-3}	–
Mask ratio	–	0.30
Projector dimensions	$512 \to 128$	–
Probe epochs	40	
Probe batch size	8	
Probe learning rate	1×10^{-3}	

7.5 Results

The adapted models are evaluated on clean and reconstructed point clouds. Robustness to reconstruction artifacts is assessed primarily on the reconstructed point clouds.

TABLE V: Comparison of supervised and self-supervised PTv3 models on Ribes_04. Δ denotes the mIoU difference (percentage points) relative to the supervised PTv3 baseline.

Method	GT mIoU (%)	Δ (pp)	REC mIoU (%)	Δ (pp)
PTv3 supervised	**93.54**	–	**87.31**	–
PTv3 + Utonia	88.76	−4.78	87.81	+0.50
PTv3 + Barlow Twins	89.50	−4.04	89.03	+1.72

TABLE VI: Mean segmentation degradation under different reconstruction conditions. Lower degradation indicates better robustness.

Condition	PTv3 (%)	Utonia(%)	Barlow Twins (%)
Coordinate noise	1.22	2.24	1.33
Density	0.1	0.14	0.04
Missing points	2.17	2.33	2.22
Missing regions	2.23	6.40	2.12
Uneven density	1.22	0.14	0.41

8. Skeletonization and Phenotyping

Following 3D segmentation, the reconstructed plant is processed for skeletonization and phenotypic trait quantification.

8.1 Laplacian-Based Skeletonization

The plant point cloud is skeletonized using Laplacian-Based Contraction (LBC) with the PC-Skeletor framework [17]. The point cloud is iteratively contracted by solving

$$\begin{bmatrix} \mathbf{W}_L \mathbf{L} \\ \mathbf{W}_H \end{bmatrix} \mathbf{P}' = \begin{bmatrix} \mathbf{0} \\ \mathbf{W}_H \mathbf{P} \end{bmatrix}, \tag{14}$$

where $\mathbf{P} \in \mathbb{R}^{n\times 3}$ is the original point cloud of n points, $\mathbf{P}' \in \mathbb{R}^{n\times 3}$ is the contracted point cloud, $\mathbf{L} \in \mathbb{R}^{n\times n}$ is the discrete Laplacian operator, and $\mathbf{W}_L$ and $\mathbf{W}_H$ are diagonal weight matrices controlling contraction and attraction, respectively. PC-Skeletor was configured with a down-sampling resolution of 0.1, 150 maximum iterations, a contraction amplification factor of 1.1, and a termination ratio of 0.0005.

Fig. 9: Skeleton extracted from the plant point cloud using Laplacian-Based Contraction.

8.2 Phenotypic Trait Quantification

The vegetation fraction is computed as

$$F_{\text{veg}} = \frac{N_{\text{vegetation}}}{N_{\text{total}}} \times 100, \tag{15}$$

where $N_{\text{vegetation}}$ and N_{total} denote the numbers of vegetation and total reconstructed points, respectively. For Ribes_04, N_{total} = 1,190,366 and $N_{\text{vegetation}}$ = 1,050,461, yielding $F_{\text{veg}} = 88.25\%$. The reconstructed plant contains 33 segmented instances, with a height of 34.64 cm and canopy widths of 21.71 cm and 22.16 cm. The bounding-box and convex-hull volumes are 16,666.63 cm^3 and 7,324.70 cm^3, respectively.

8.3 Plant Health Estimation

RGB-based plant health classification divides the reconstructed points into healthy, moderate, stressed/chlorotic, and necrotic/senescent categories.

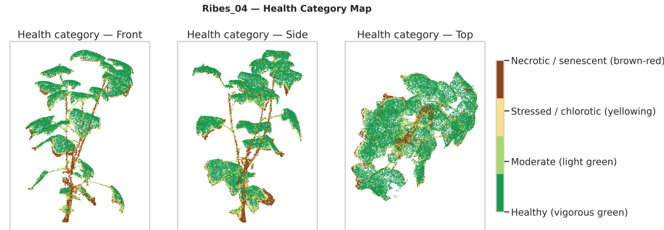


Fig. 10: Color-based plant health classification.

9. Conclusion

This paper presented a monocular RGB robotic pipeline for automated 3D plant phenotyping. To overcome reconstruction-induced segmentation degradation, self-supervised feature adaptation using Barlow Twins improved segmentation mIoU on reconstructed point clouds by +1.72 percentage points. Paired with skeletonization and color analysis, the system successfully automates the extraction of key structural traits—such as height, volume, and plant health—proving the viability of monocular RGB imaging for robust 3D phenotyping.

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